SSS 2 CHEMISTRY – MULTIPLE-CHOICE QUESTIONS

Instruction: Choose the correct answer from options A–D.

1. What is the relative atomic mass of an element?

A. The mass of one proton
B. The average mass of its atoms compared with 1/12 of carbon-12
C. The number of electrons in the atom
D. The number of neutrons in the atom
Answer: B

2. What is the relative molecular mass of water, H_2O?

(H = 1, O = 16)
A. 16
B. 17
C. 18
D. 20
Answer: C

3. What is the relative molecular mass of CO_2?

(C = 12, O = 16)
A. 28
B. 32
C. 40
D. 44
Answer: D

4. How many moles are present in 18 g of water?

(H = 1, O = 16)
A. 0.5 mol
B. 1 mol
C. 2 mol
D. 18 mol
Answer: B

5. What is the molar mass of sodium hydroxide, NaOH?

(Na = 23, O = 16, H = 1)
A. 30 g/mol
B. 39 g/mol

C. 40 g/mol
D. 41 g/mol
Answer: C

6. One mole of a substance contains approximately?

A. 6.02×10^{23} particles
B. 3.01×10^{23} particles
C. 1.00×10^{23} particles
D. 9.81×10^{23} particles
Answer: A

7. The number 6.02×10^{23} is known as?

A. Atomic number
B. Avogadro constant
C. Gas constant
D. Faraday constant
Answer: B

8. What is the empirical formula of a compound?

A. The formula showing the actual number of atoms in a molecule
B. The simplest whole-number ratio of atoms in a compound
C. The formula showing only metals
D. The formula showing only non-metals
Answer: B

9. What is the empirical formula of benzene, C_6H_6?

A. CH
B. CH_2
C. C_2H_2
D. C_6H_6
Answer: A

10. Which law states that matter is neither created nor destroyed during a chemical reaction?

A. Law of multiple proportions
B. Law of conservation of mass
C. Boyle's law

D. Charles' law
Answer: B

11. A chemical equation is said to be balanced when?

A. The number of molecules is equal on both sides
B. The total mass of reactants equals the total mass of products
C. The reactants are heavier than the products
D. The products are heavier than the reactants
Answer: B

12. Which of the following is a balanced equation?

A. $H_2 + O_2 \rightarrow H_2O$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $H_2 + 2O_2 \rightarrow H_2O$
D. $2H_2 + 2O_2 \rightarrow H_2O$
Answer: B

13. What type of reaction occurs when two or more substances combine to form one product?

A. Decomposition
B. Combination
C. Displacement
D. Neutralisation
Answer: B

14. What type of reaction is represented by AB → A + B?

A. Combination
B. Decomposition
C. Neutralisation
D. Displacement
Answer: B

15. Which of the following is an example of a displacement reaction?

A. $HCl + NaOH \rightarrow NaCl + H_2O$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
D. $H_2 + Cl_2 \rightarrow 2HCl$
Answer: C

16. An acid is a substance that produces which ion in aqueous solution?

A. OH^-
B. H^+
C. Na^+
D. Cl^-
Answer: B

17. A base is a substance that produces which ion in aqueous solution?

A. H^+
B. Cl^-
C. OH^-
D. Na^+
Answer: C

18. Which of the following is a strong acid?

A. Ethanoic acid
B. Carbonic acid
C. Hydrochloric acid
D. Citric acid
Answer: C

19. Which of the following is a weak acid?

A. Hydrochloric acid
B. Sulfuric acid
C. Nitric acid
D. Ethanoic acid
Answer: D

20. What is the pH of a neutral solution at room temperature?

A. 0
B. 5
C. 7
D. 14
Answer: C

21. A solution with a pH of 3 is?

A. Strongly alkaline
B. Weakly alkaline

C. Neutral
D. Acidic
Answer: D

22. A solution with a pH of 11 is?

A. Acidic
B. Alkaline
C. Neutral
D. Saturated
Answer: B

23. Which indicator is commonly used to test whether a solution is acidic or alkaline?

A. Litmus
B. Starch
C. Copper sulfate
D. Sodium chloride
Answer: A

24. What colour does blue litmus paper turn in an acidic solution?

A. Blue
B. Green
C. Red
D. Yellow
Answer: C

25. What colour does red litmus paper turn in an alkaline solution?

A. Blue
B. Red
C. Yellow
D. Orange
Answer: A

26. A reaction between an acid and a base is called?

A. Oxidation
B. Neutralisation
C. Reduction
D. Polymerisation
Answer: B

27. Which of the following is a product of neutralisation?

A. Salt and water
B. Hydrogen and oxygen
C. Carbon and water
D. Metal and oxygen
Answer: A

28. Which salt is formed when hydrochloric acid reacts with sodium hydroxide?

A. Sodium sulfate
B. Sodium chloride
C. Sodium carbonate
D. Sodium nitrate
Answer: B

29. What is the process of obtaining pure crystals of a soluble solid from its solution called?

A. Filtration
B. Crystallisation
C. Decantation
D. Sublimation
Answer: B

30. What is the concentration of a solution containing 1 mole of solute in 1 dm^3 of solution?

A. 0.1 mol/dm^3
B. 0.5 mol/dm^3
C. 1.0 mol/dm^3
D. 10 mol/dm^3
Answer: C

31. Which apparatus is commonly used to measure a fixed volume of solution accurately?

A. Pipette
B. Beaker
C. Evaporating dish
D. Test tube
Answer: A

32. Which apparatus is used to deliver a measured variable volume of solution during titration?

A. Measuring cylinder
B. Burette
C. Funnel
D. Balance
Answer: B

33. The point at which an indicator changes colour during titration is called the?

A. Melting point
B. End point
C. Boiling point
D. Freezing point
Answer: B

34. Which of the following gases is produced when zinc reacts with dilute hydrochloric acid?

A. Oxygen
B. Nitrogen
C. Hydrogen
D. Carbon dioxide
Answer: C

35. Which test is used to identify hydrogen gas?

A. It relights a glowing splint
B. It produces a pop sound with a lighted splint
C. It turns limewater milky
D. It turns blue litmus red
Answer: B

36. Which gas turns limewater milky?

A. Oxygen
B. Hydrogen
C. Nitrogen
D. Carbon dioxide
Answer: D

37. Which gas supports combustion?

A. Oxygen
B. Carbon dioxide
C. Nitrogen
D. Hydrogen
Answer: A

38. What is oxidation?

A. Gain of electrons only
B. Loss of electrons or gain of oxygen
C. Loss of oxygen only
D. Gain of hydrogen only
Answer: B

39. What is reduction?

A. Gain of oxygen
B. Loss of electrons
C. Gain of electrons or loss of oxygen
D. Loss of hydrogen only
Answer: C

40. A substance that causes another substance to be oxidised is called?

A. Reducing agent
B. Oxidising agent
C. Catalyst
D. Indicator
Answer: B

41. A catalyst is a substance that?

A. Stops a chemical reaction completely
B. Increases the rate of reaction without being permanently consumed
C. Changes the products of a reaction
D. Always decreases reaction rate
Answer: B

42. Which factor generally increases the rate of a chemical reaction?

A. Decreasing temperature
B. Decreasing concentration
C. Increasing temperature

D. Removing reactants
Answer: C

43. Increasing the surface area of a solid reactant generally causes the reaction rate to?

A. Increase
B. Decrease
C. Remain zero
D. Stop completely
Answer: A

44. According to Boyle's law, at constant temperature, the pressure of a fixed mass of gas is?

A. Directly proportional to its volume
B. Inversely proportional to its volume
C. Equal to its mass
D. Independent of volume
Answer: B

45. According to Charles' law, at constant pressure, the volume of a fixed mass of gas is directly proportional to its?

A. Pressure
B. Density
C. Absolute temperature
D. Mass
Answer: C

46. What happens to the volume of a gas when its temperature increases at constant pressure?

A. It decreases
B. It increases
C. It becomes zero
D. It remains unchanged
Answer: B

47. Organic Chemistry is mainly the study of compounds of?

A. Sodium
B. Carbon

C. Calcium
D. Iron
Answer: B

48. Which of the following is an alkane?

A. Ethene
B. Ethyne
C. Ethane
D. Ethanol
Answer: C

49. What is the general formula of alkanes?

A. C_nH_{2n}
B. C_nH_{2n+2}
C. C_nH_{2n-2}
D. C_nH_n
Answer: B

50. Which of the following is the correct molecular formula for methane?

A. CH_4
B. C_2H_4
C. C_2H_6
D. CH_3OH
Answer: A